



中國人民大學
RENMIN UNIVERSITY OF CHINA



中国人民大学信息学院
SCHOOL OF INFORMATION RENMIN UNIVERSITY OF CHINA



DeepAnalyze: 自主数据科学初探

张绍磊

中国人民大学

2026.03.18

个人信息



张绍磊 / 中国人民大学信息学院 助理教授

研究
方向

- 人工智能（大模型、Agentic）
- 数据科学（数据治理、数据分析）

大模型项目在GitHub社区获得广泛关注，累积获得 **10000+** 星标

ictnlp/BayLing Public

“百聆”是一个基于LLaMA的语言对齐增强的英语/中文大语言模型，具有优越的英语/中文能力，在多语言和通用任务等多项测试中取得ChatGPT 90%的性能。BayLing is an English/Chinese LLM equipped with advanced language alignment, showing superior capability in English/Ch...

Python 318 20

ictnlp/StreamSpeech

Public

StreamSpeech is an “All in One” seamless model for offline and simultaneous speech recognition, speech translation and speech synthesis.

Python 1.2k 98

LLaMA-Omni Public

LLaMA-Omni is a low-latency and high-quality end-to-end speech interaction model built upon Llama-3.1-8B-Instruct, aiming to achieve speech capabilities at the GPT-4o level.

Python 3.1k 216

LLaVA-Mini Public

LLaVA-Mini is a unified large multimodal model (LMM) that can support the understanding of images, high-resolution images, and videos in an efficient manner.

Python 543 28

ictnlp/Stream-Omni Public

Stream-Omni is a GPT-4o-like language-vision-speech chatbot that simultaneously supports interaction across various modality combinations.

Python 355 38

ruc-datalab/DeepAnalyze

Public

DeepAnalyze is the first agentic LLM for autonomous data science. ❤️ 你的AI数据分析师，自动分析大量数据，一键生成专业分析报告！

Python 2.3k 325

BayLing
318 Stars

StreamSpeech
1200+ Stars

LLaMA-Omni
3100+ Stars

LLaVA-Mini
548 Stars

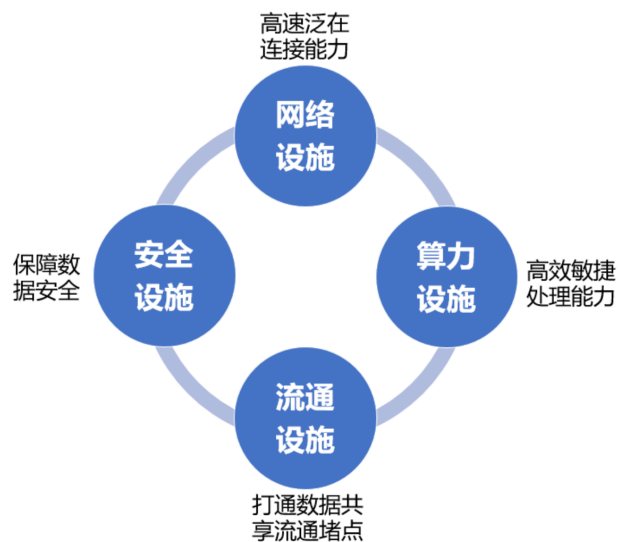
Stream-Omni
385 Stars

DeepAnalyze
3800+ Stars

数据科学是释放数据价值的关键

数据科学 提升数据科学系统的智能化水平与性能，高效释放数据价值

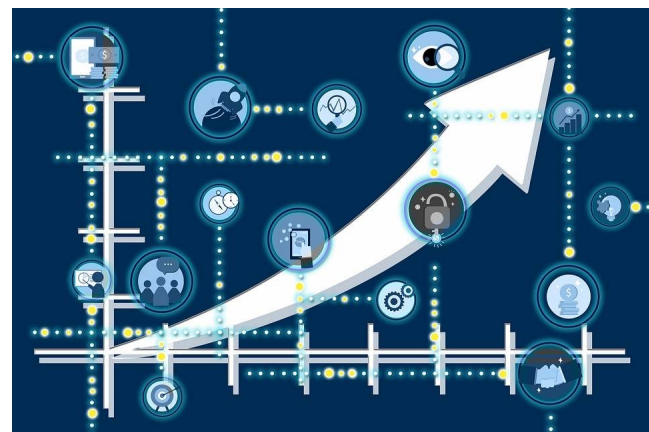
助力数据基础设施建设



促进智慧城市精细治理



推动产业数智化转型



数据科学技术起到了重要的基础性支撑作用



来源：《国家数据局数据技术设施体系》《数据空间发展战略蓝皮书》

什么是数据科学？

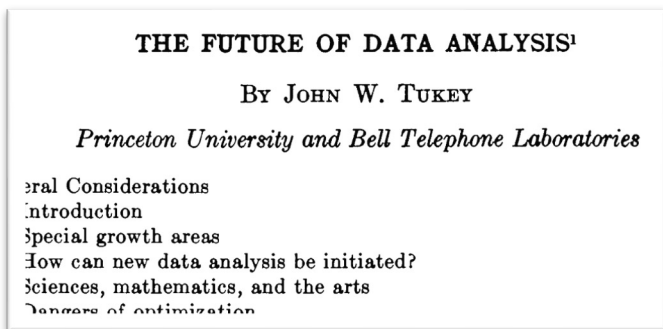


数据技术的发展历程

数据技术

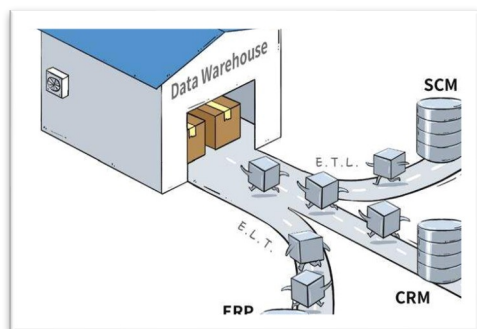
从数据库技术到大数据技术

数据来源 ↗ 数据规模 ↗ 数据质量 ↗



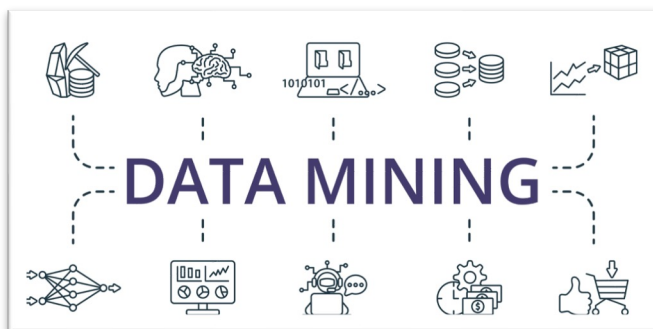
数据科学概念被提出

1962



关系数据库、数据仓库

1960s-1990s



数据挖掘

2000s



大数据

2010s

智能技术的发展历程

智能技术

从机器学习到深度学习

模态数量



模型能力

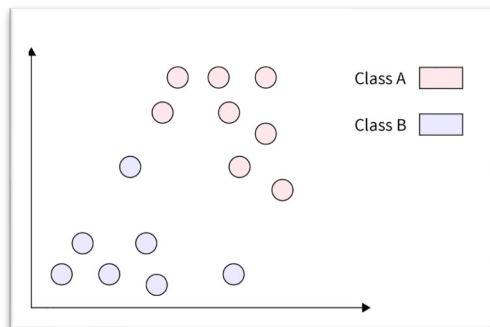


任务类型



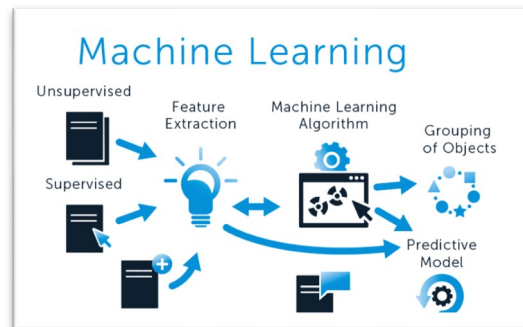
人工智能概念被提出

1956



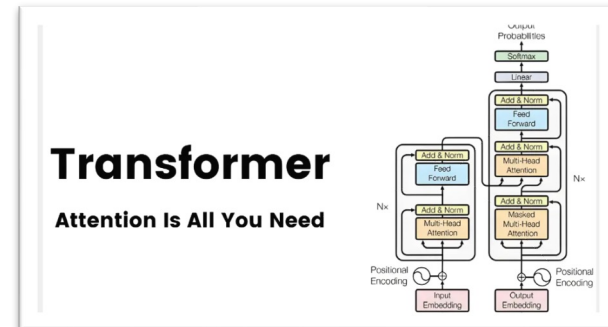
专家系统、统计学习

1960s-1990s



机器学习

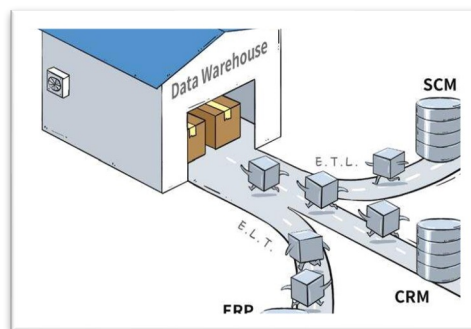
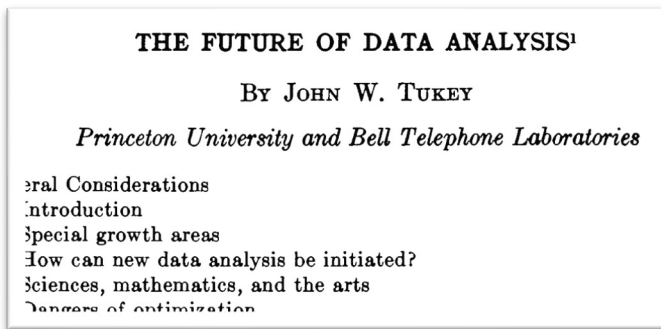
2000s



深度学习

2010s

数据技术的发展历程



数据科学概念被提出

关系数据库、数据仓库

数据挖掘

大数据

~1960

1960s-1990s

2000s

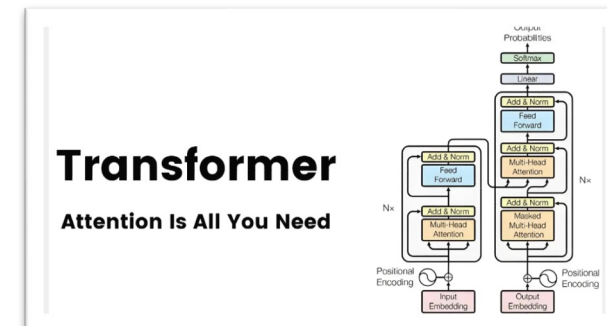
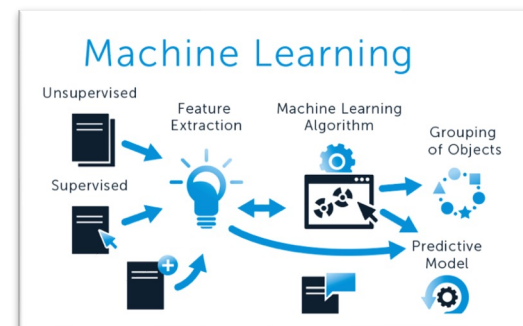
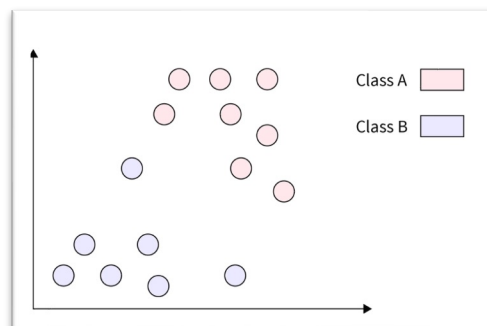
2010s

人工智能概念被提出

专家系统、统计学习

机器学习

深度学习



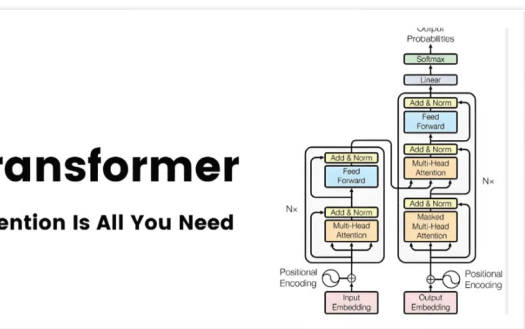
数据 + 智能



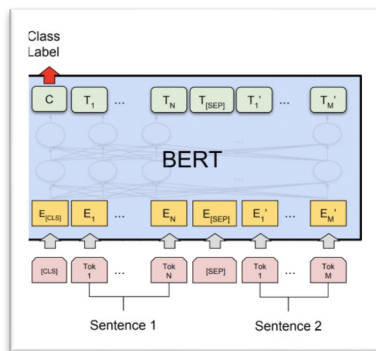
大数据

2010s

深度学习



完成数据准备、分析
等少量数据科学任务



2019

BERT/GPT等
预训练模型出现

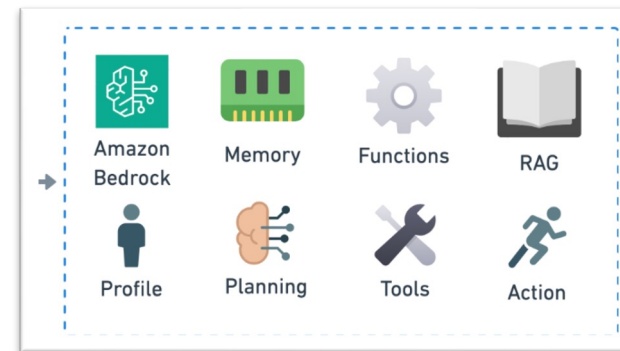
通过Prompt工程完成
单一数据科学任务



2022

ChatGPT等
大模型（LLM）出现

通过Agent完成复杂
数据科学Pipeline



2024

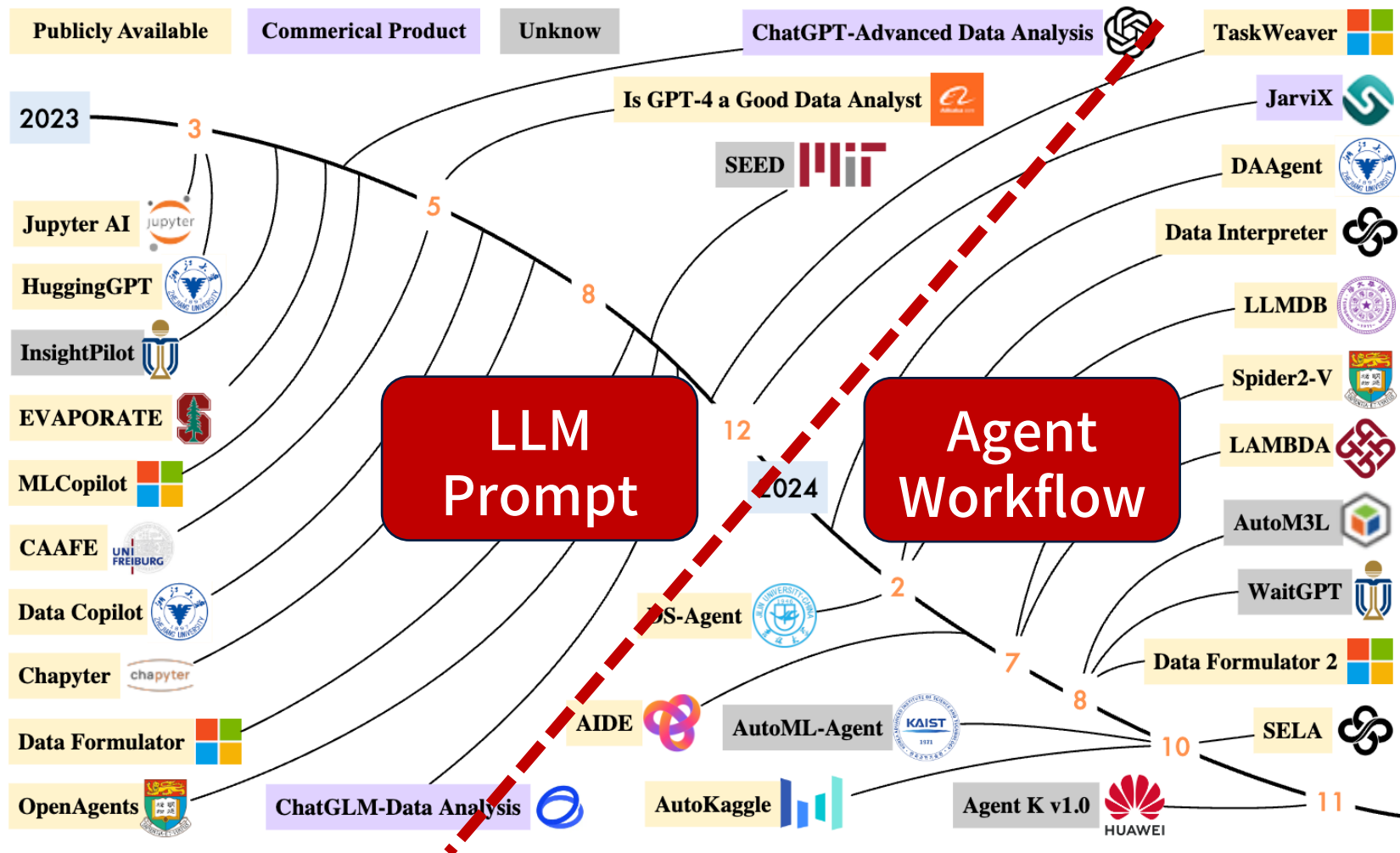
Agent
智能体出现

数据科学智能体

AI for Data Science: 基于大模型的数据科学系统



以大模型为核心的数据科学系统成为研究热点



为什么大模型完成数据科学任务成为趋势？

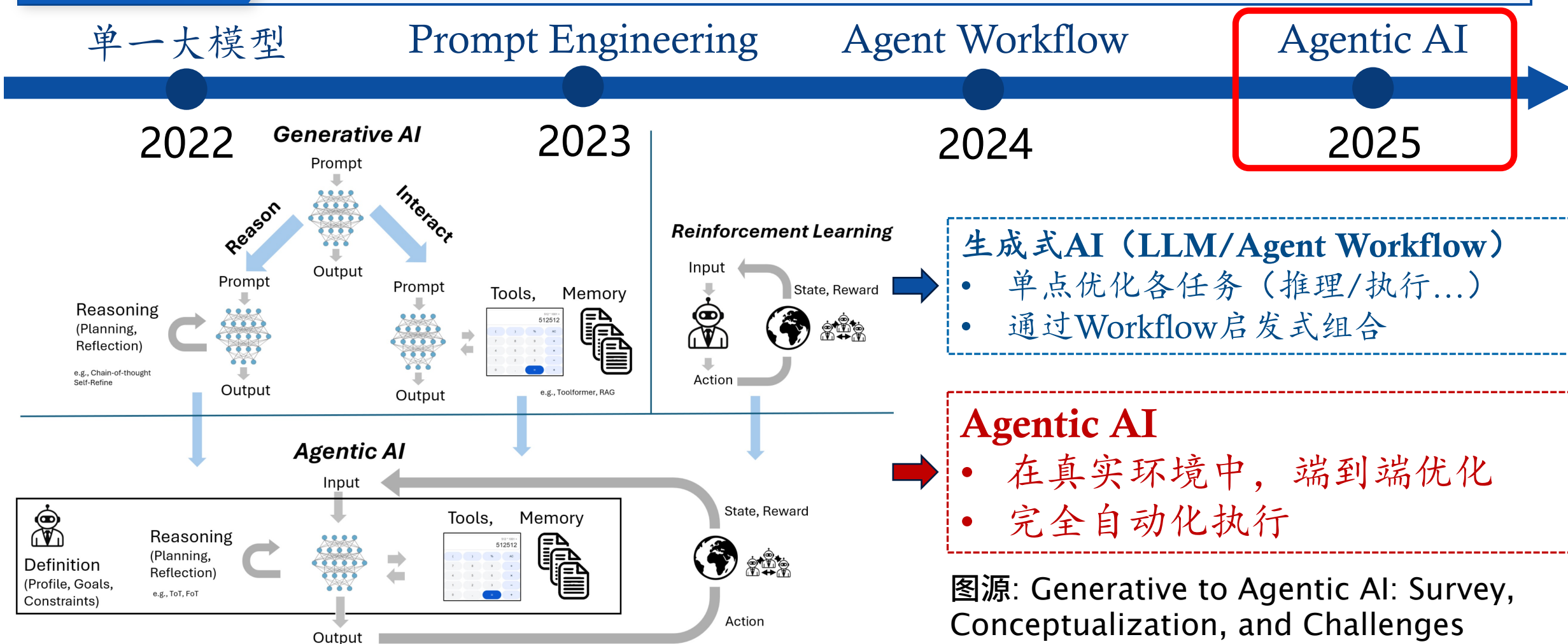
- 数据科学任务在各领域需求丰富，亟需解决
- 数据科学任务难，直接反应LLM系统的智能化水平

图源: A Survey on Large Language Model-based Agents for Statistics and Data Science

AI for Data Science: Agentic数据科学系统



Agentic AI 大模型是否能自主完成数据科学任务?



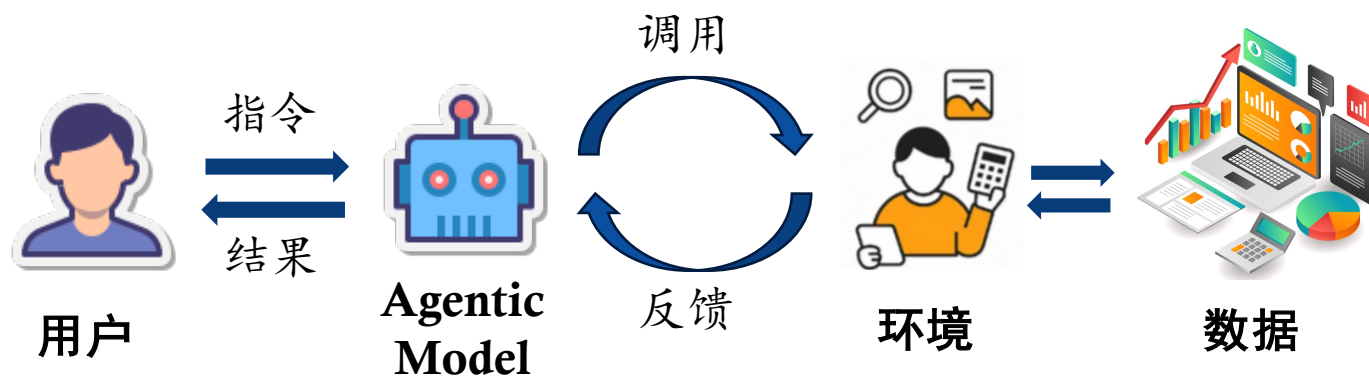
图源: Generative to Agentic AI: Survey, Conceptualization, and Challenges

AI for Data Science: Agentic数据科学系统



Agentic AI

大模型是否能自主完成数据科学任务？



Agentic 数据科学系统特征：

1. 声明式交互
2. 自动化与环境交互
3. 全流程联合优化
4. 多模态感知

DeepAnalyze: 自主数据科学大模型

DeepAnalyze 模拟数据科学家的行为，自动从数据源中提取insight

DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

Shaolei Zhang¹ Ju Fan¹ Meihao Fan¹ Guoliang Li² Xiaoyong Du¹

 [ruc-datalab/DeepAnalyze](https://github.com/ruc-datalab/DeepAnalyze)  DeepAnalyze-8B  DataScience-Instruct-500K  [ruc-deepanalyze.github.io](https://github.com/ruc-deepanalyze/ruc-deepanalyze.github.io)



能胜任开放域下数据科学任务：

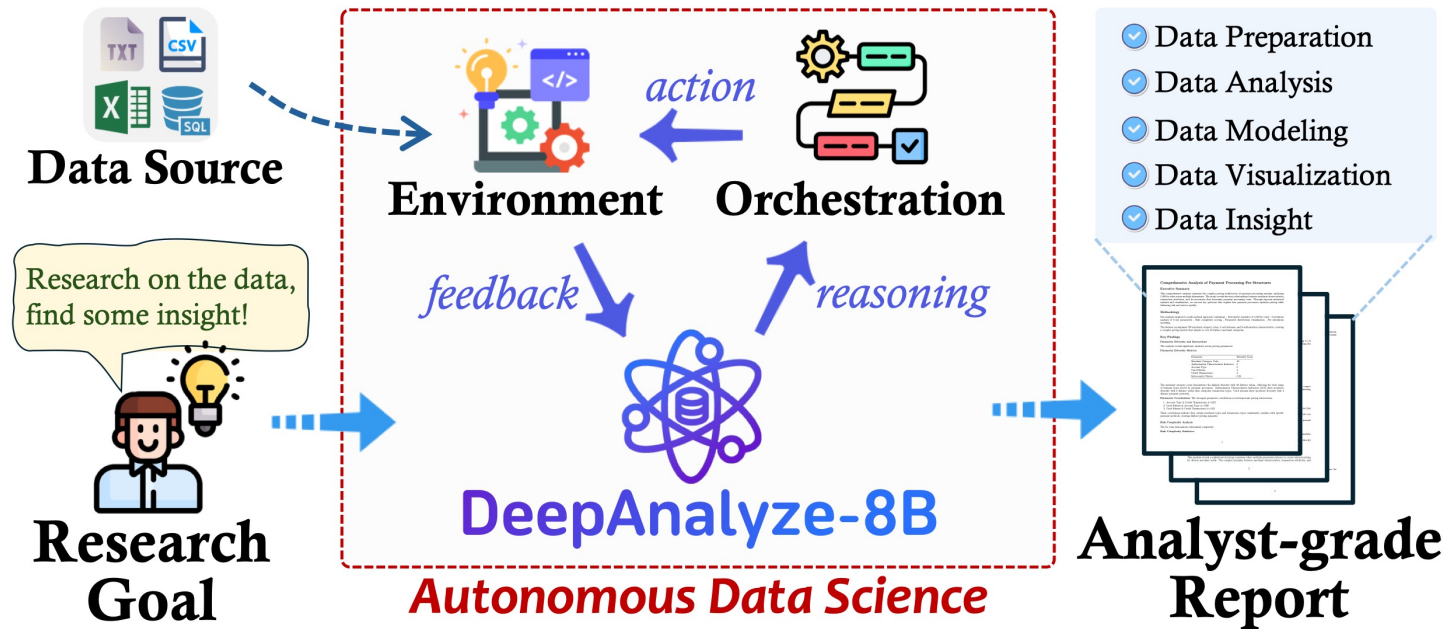
1. 分析任务，并规划步骤
2. 在环境中读取数据
3. 观察并理解数据
4. 通过代码完成数据分析、建模、可视化...
5. 总结、呈现结果

DeepAnalyze: 自主数据科学大模型



DeepAnalyze

模拟数据科学家的行为，自动从数据源中提取insight



给定数据源，DeepAnalyze自主编排操作、根据反馈自适应优化

- **全流程数据科学Pipeline**：数据准备、分析、建模、可视化、洞察
- **开放式数据研究**：自动研究数据，生成分析师级别报告

DeepAnalyze: 自主数据科学大模型

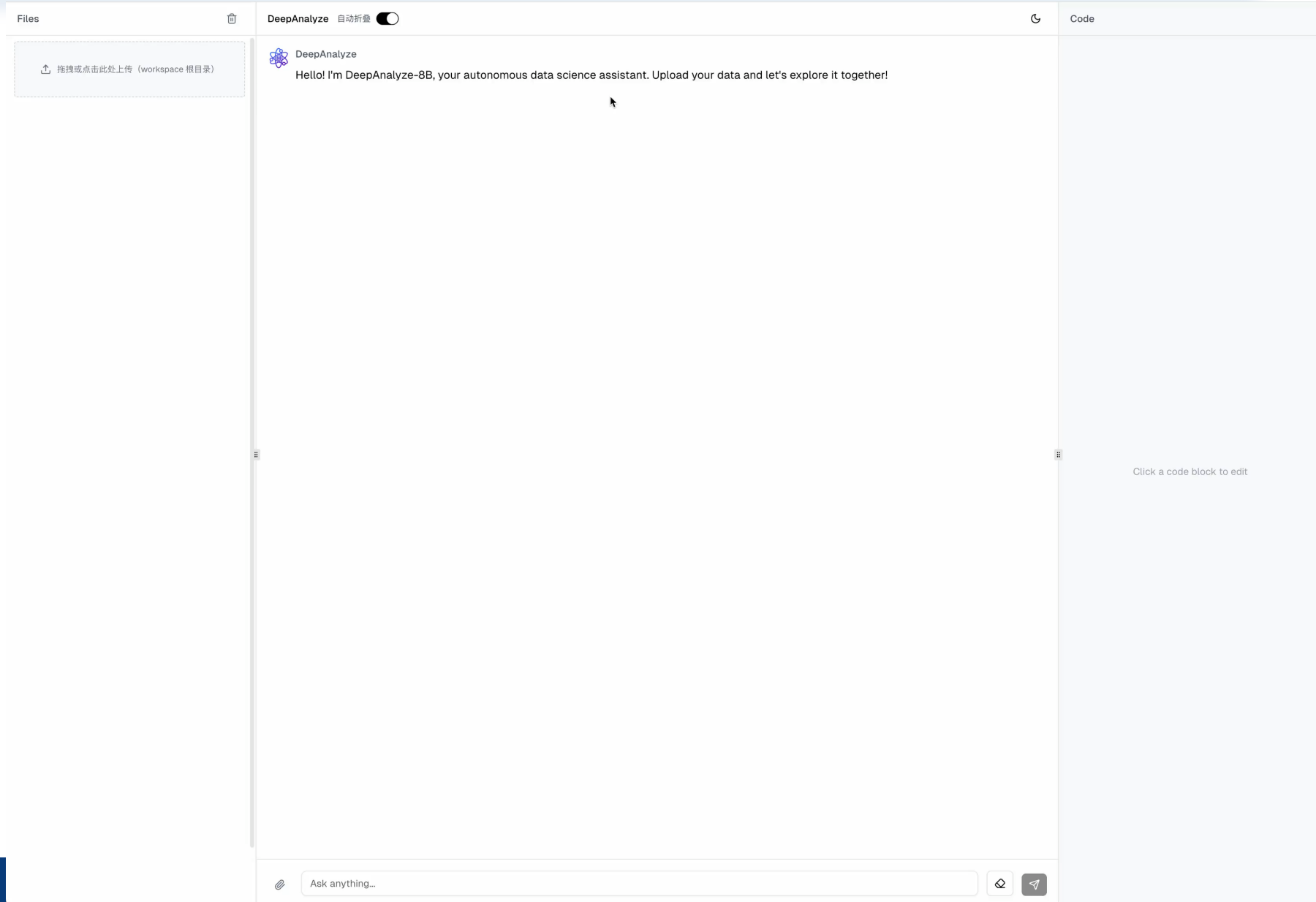


DeepAnalyze-8B

- (1) 处理多种类型数据
- (2) 自动编排+自适应优化
- (3) 开放式数据研究，生成分析师级别研究报告



开源Code/Demo
3800+ stars



DeepAnalyze: 自主数据科学大模型



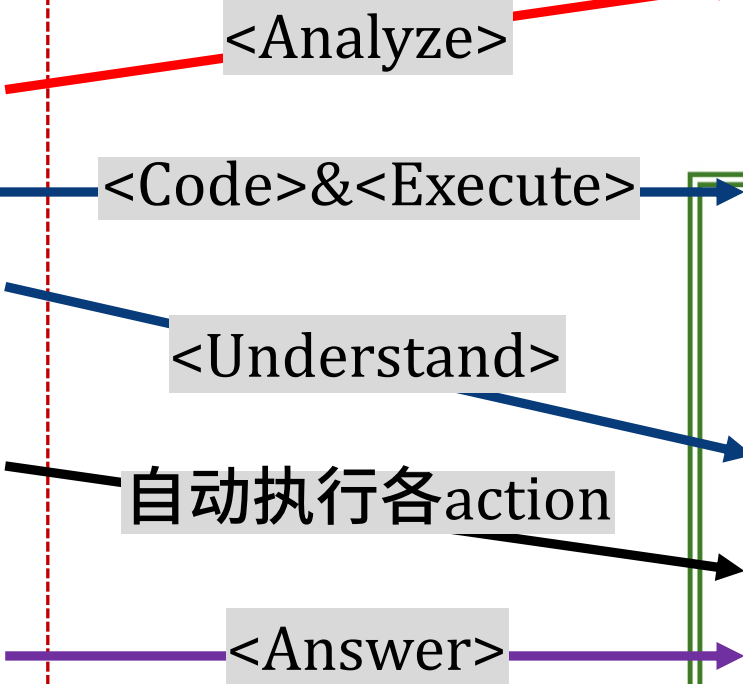
引入若干action，使LLM能与数据交互

数据科学家

DeepAnalyze

完成数据科学任务的步骤:

- 1. 分析任务，并规划步骤
- 2. 在环境中读取数据
- 3. 观察并理解数据
- 4. 通过代码完成数据分析、建模、可视化...重复直到完成任务
- 5. 总结、呈现结果



Data:

Instruction:
Give me a comprehensive data analysis report based on these files.
Requirements: (1)...(2)...

Response:

<Analyze>
To complete this task, I need to (1) load the data, (2) observe its structure, (3) select a statistical model, ... (8) finally generate an analysis report. Next, I will first look at the data.

</Analyze>

<Code>
python
import pandas as pd
df = pd.read_csv('data.csv')
print("=== First 5 Rows ===\n", df.head()) ...

</Code>

<Execute>
=== First 5 Rows ===
Name Age Salary Department
Bob 23 6,000 Marketing ...

</Execute>

<Understand>
Through the execution results, I know the csv file is about ..., it has 4 columns...

</Understand>

<Analyze>...<Understand>...<Code><Execute>...

<Answer>
Analysis Report of Data:...

</Answer>

Intermediate Results

- Processed Data
- Statistical Analysis
- Data Insight
- Visualization
- Data Modeling

Real-world Environment

DeepAnalyze: 自主数据科学大模型



引入若干action，使LLM能与数据交互

数据科学家

DeepAnalyze

完成数据科学任务的步骤：

- 1.
- 2.
- 3.
4. 通过代码完成数据分析、建模、可视化...重复直到完成任务
5. 总结、呈现结果

通过action定义，DeepAnalyze能模仿数据科学家与数据交互

如何学习数据科学思维、流程、能力？

<Understand>

自动执行各action

<Answer>

Data:



Instruction:

Give me a comprehensive data analysis report based on these files.
Requirements: (1)...(2)...

Response:

<Analyze>

To complete this task, I need to (1) load the data, (2) observe its structure, (3) select a statistical model, ... (8) finally generate an analysis report.

Intermediate Results



Processed Data



Statistical Analysis



Data Insight



Visualization



Data Modeling

=== First 5 Rows ===

Name	Age	Salary	Department
Bob	23	6,000	Marketing ...

</Execute>

<Understand>

Through the execution results, I know the csv file is about ..., it has 4 columns...

</ Understand >

<Analyze>...<Understand>...<Code><Execute>...

<Answer>

Analysis Report of Data:...

</Answer>

Real-world Environment

DeepAnalyze: 自主数据科学大模型



- **Agentic Training: 在真实环境中通过强化学习训练LLM**
 - 在深度搜索等领域已经成功实践
- **在数据科学任务上的局限性:**
 - **奖励稀疏:**
 - 数据科学任务复杂 → LLM无法完成任务 → 正向奖励稀疏, LLM无法训练
 - **轨迹稀缺:**
 - 完成任务轨迹空间大 → 缺少正确的推理/交互轨迹 → LLM陷入盲目试错

DeepAnalyze: 自主数据科学大模型



- **Agentic Training: 在真实环境中通过强化学习训练LLM**
 - 在深度搜索等领域已经成功实践

- **在数据科学任务上的局限性:**

- **奖励稀疏:**

课程式Agentic Training

- 数据科学任务复杂 → LLM无法完成任务 → 正向奖励稀疏, LLM无法训练

- **轨迹稀缺:**

Agentic 数据合成

- 完成任务轨迹空间大 → 缺少正确的推理/交互轨迹 → LLM陷入盲目试错

DeepAnalyze: 自主数据科学大模型



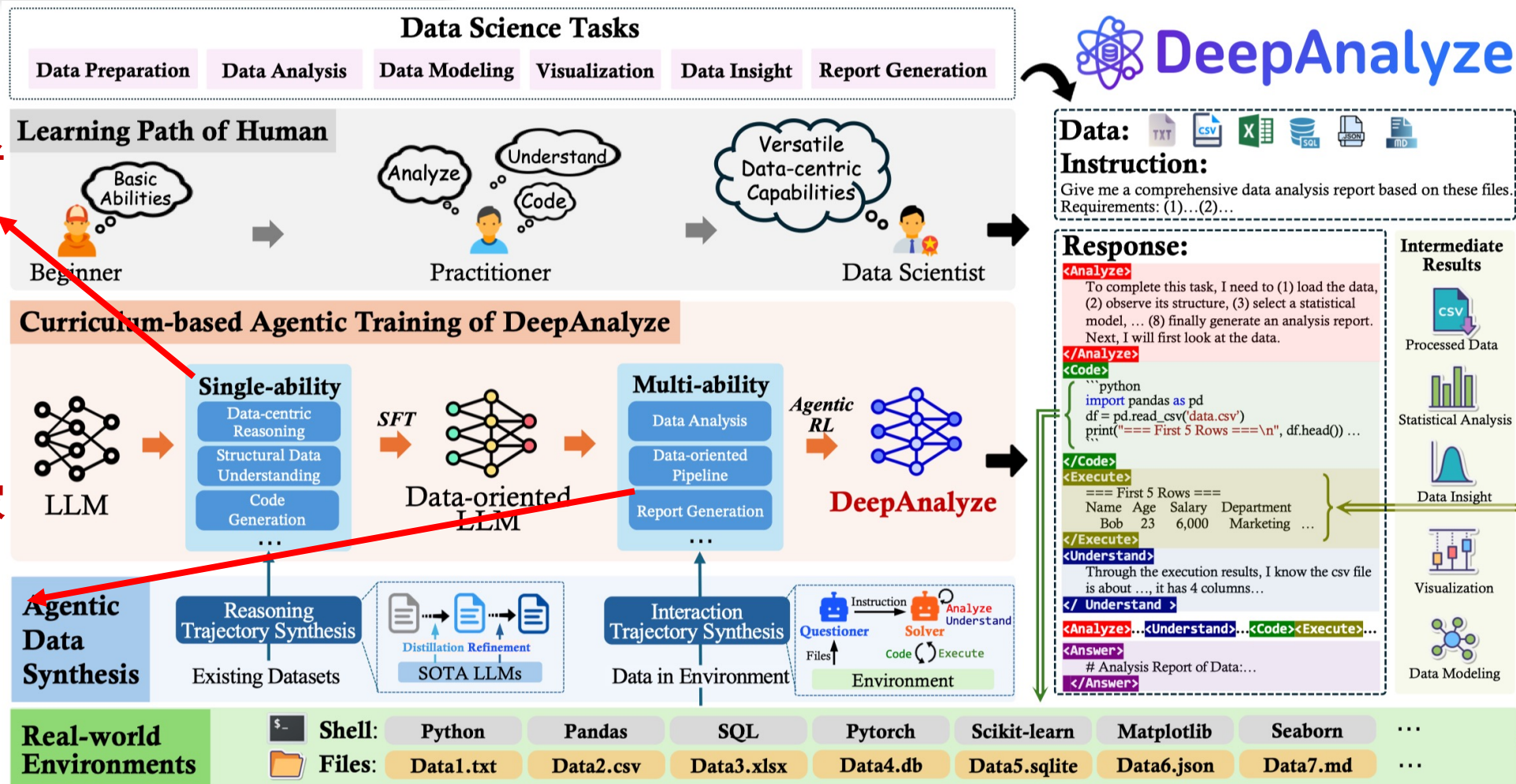
奖励稀疏：课程式Agentic Training 由单一能力到复合能力逐步学习

阶段1：初学者→从业者

- 学习推理、结构化数据理解、代码等单一能力

阶段2：从业者→科学家

- 学习多能力的复杂任务
- 真实环境中，自我进化

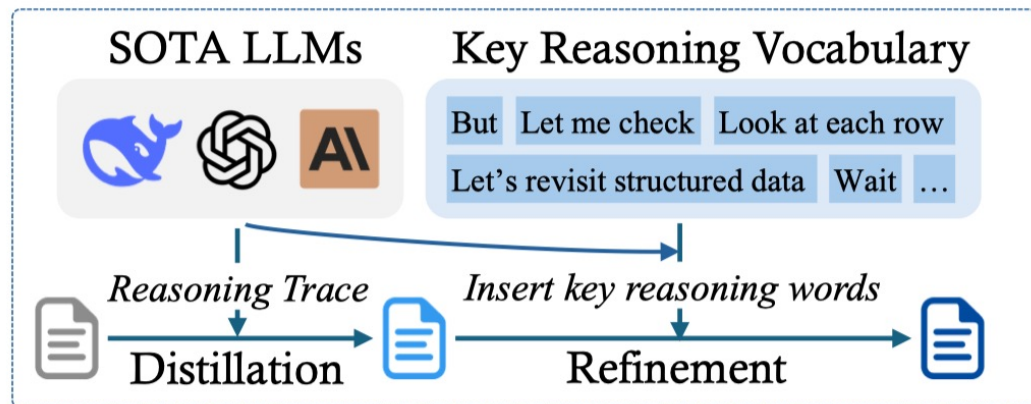


DeepAnalyze: 自主数据科学大模型

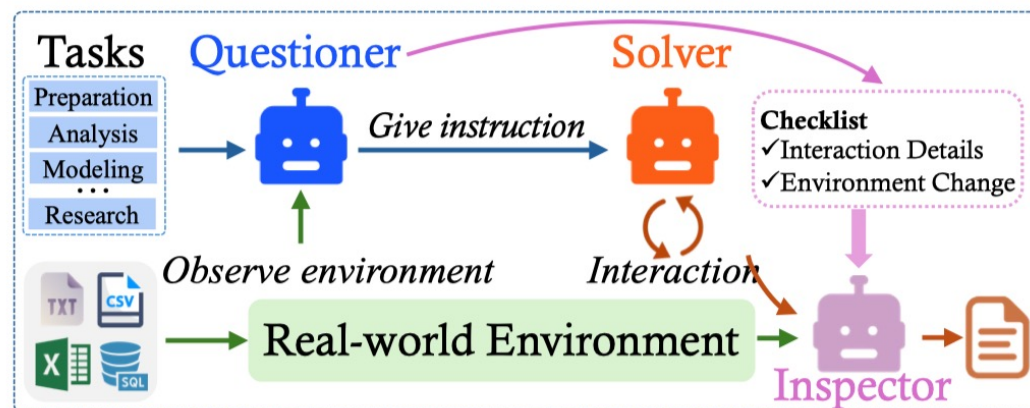


轨迹稀缺：面向数据的轨迹合成框架

推理轨迹+交互轨迹，共500K数据



(a) Reasoning Trajectory Synthesis



(b) Interaction Trajectory Synthesis

推理轨迹合成

- ✓ 蒸馏SOTA闭源模型
- ✓ 在推理轨迹中插入关键推理词汇（But/Wait/Look at each row），增强推理轨迹对数据的关注

交互轨迹合成

- ✓ Questioner：根据数据提问题
- ✓ Solver：和环境交互完成任务
- ✓ Inspector：检测交互轨迹是否合法

DeepAnalyze: 自主数据科学大模型



DeepAnalyze 在DataSciBench展现出全流程数据科学能力

Models	Coarse-grained Metrics		Fine-grained Metrics						Score
	Success Rate	Completion Rate	VLM	F1: Data Preparation	F2: Plot Validity	F3: Data Exploration	F4: Data Visualization	F5: Data Modeling	
Close-Source API-Based Agent									
o1-mini	29.77	45.26	2.87	44.63	19.27	36.01	30.94	23.81	38.78
GPT-4o-mini	50.63	57.78	3.05	60.30	48.02	57.84	59.24	53.54	54.18
GPT-4o	66.31	68.44	3.91	75.93	56.14	69.33	71.35	57.67	64.51
GPT-4-Turbo	51.93	58.87	3.09	62.30	41.62	57.75	60.25	50.75	54.65
Claude-3-5-Sonnet	47.48	58.11	2.14	49.07	36.94	55.84	52.87	46.04	52.29
GLM-4-Flash	30.32	34.04	1.33	36.53	29.42	32.57	27.64	14.44	30.74
Open-Source LLM-based Agent									
Llama-3.1-8B-Instruct	24.73	33.89	1.29	38.24	18.25	21.98	22.89	25.85	29.69
Gemma-2-9B-it	7.07	11.00	1.06	26.16	16.90	23.81	18.11	17.15	12.66
GLM-4-9B-Chat	25.72	30.38	1.69	31.51	23.15	28.07	27.19	19.14	27.57
Qwen2.5-7B-Instruct	43.83	50.74	1.43	51.18	36.41	47.25	45.24	34.77	45.99
Qwen2-7B-Instruct	22.84	25.58	1.16	30.93	20.78	28.73	25.87	7.52	23.52
Yi-1.5-9B-Chat-16K	38.20	42.35	0.73	38.14	36.36	35.64	37.08	27.79	38.22
CodeLlama-13B-Instruct	10.49	14.64	0.04	11.67	11.34	9.43	14.43	5.15	12.64
CodeLlama-7B-Instruct	2.88	3.97	0.00	3.53	2.37	2.57	1.74	1.59	3.31
StarCoder2-15B	2.07	2.61	0.07	2.57	1.81	1.59	3.43	1.19	2.33
Deepseek-Coder-6.7B-instruct	37.03	41.62	1.93	43.49	34.57	46.36	46.49	18.09	38.45
Qwen2.5-Coder-7B-Instruct	45.18	53.11	1.48	51.58	43.21	43.87	42.50	35.23	47.67
Agentic Model									
DeepAnalyze-8B	59.91	66.24	2.86	71.68	67.86	58.62	69.09	33.33	61.11

F1:数据准备
F2:指令跟随
F3:数据分析
F4:数据可视化
F5:数据建模

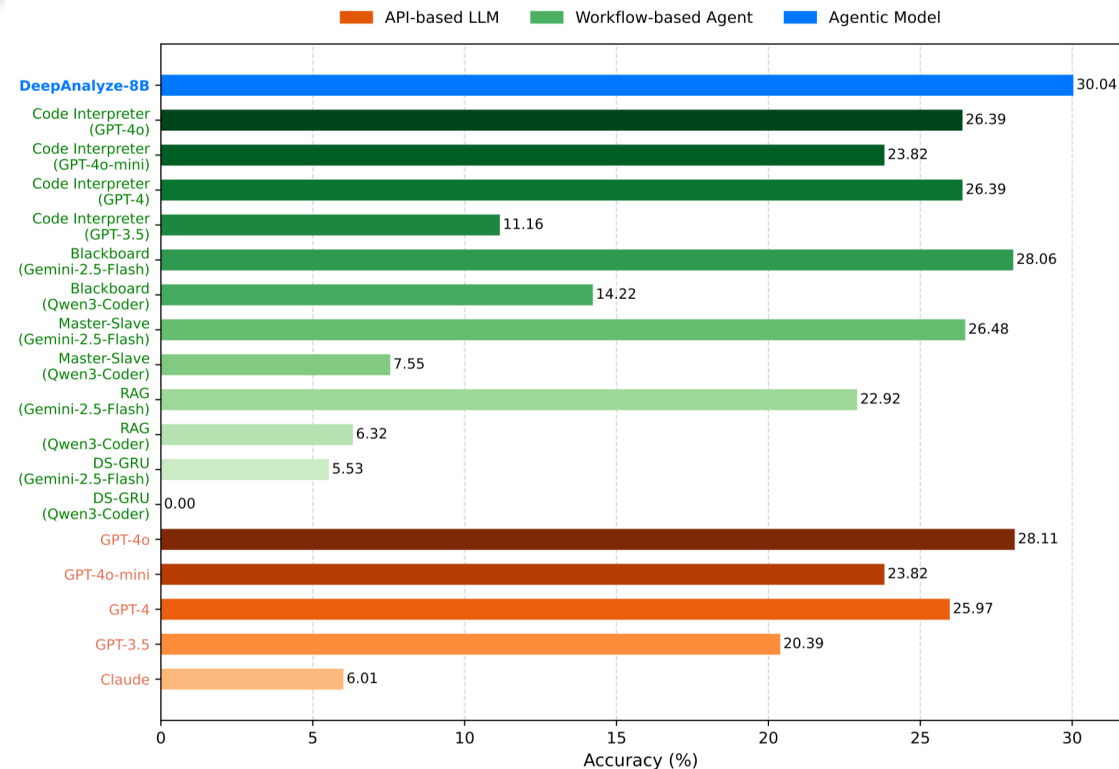
8B模型媲美
闭源模型

DeepAnalyze: 自主数据科学大模型



DeepAnalyze

数据分析、数据建模能力



数据分析

Methods	LLM	Success (%)	Performance	Cost (\$)
Workflow-based Agent				
AutoGen	Llama3-8b	5.41	1.55	0.00
	Llama3-70b	16.22	7.79	0.00
	GPT-3.5	8.11	6.02	0.41
	GPT-4	87.84	45.52	19.34
	GPT-4o	71.62	34.74	12.27
	GPT-4o-mini	22.97	11.24	0.10
Code Interpreter	GPT-3.5	16.22	6.52	2.74
	GPT-4	54.05	26.14	38.81
	GPT-4o	44.59	19.87	19.26
	GPT-4o-mini	39.19	16.90	2.70
Agentic Model				
DeepAnalyze-8B		90.63	39.41	0.00

数据建模

DeepAnalyze-8B优于workflow-based agent

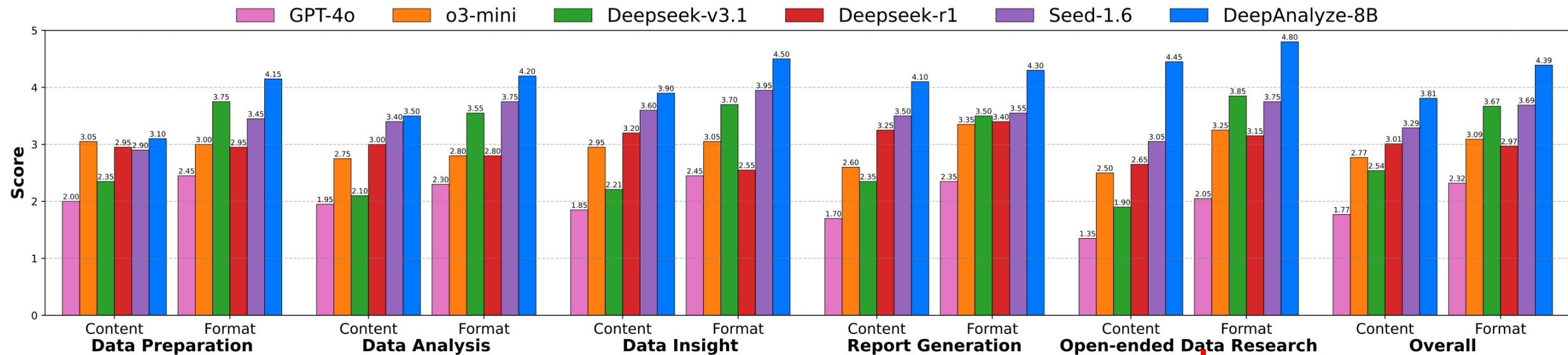
DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

DeepAnalyze: 自主数据科学大模型



DeepAnalyze 深度数据研究能力

自动生成研究报告，从内容 (content)、格式 (Format) 评分



DeepAnalyze-8B 优于闭源 LLM (Tool Call)

闭源 LLM 未经训练，在指令中没有具体约束时，无法完成开放式数据研究

DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

DeepAnalyze: 自主数据科学大模型



在HuggingFace上开源模型下载**12K+**，开源数据集下载**15K+**，曾在Huggingface日趋势榜**论文排名第一**

DeepAnalyze代码于2025年10月21日在GitHub上正式开源，已获得**3.8K+**个GitHub星标

< Papers arxiv:2510.16872 Hugging Face

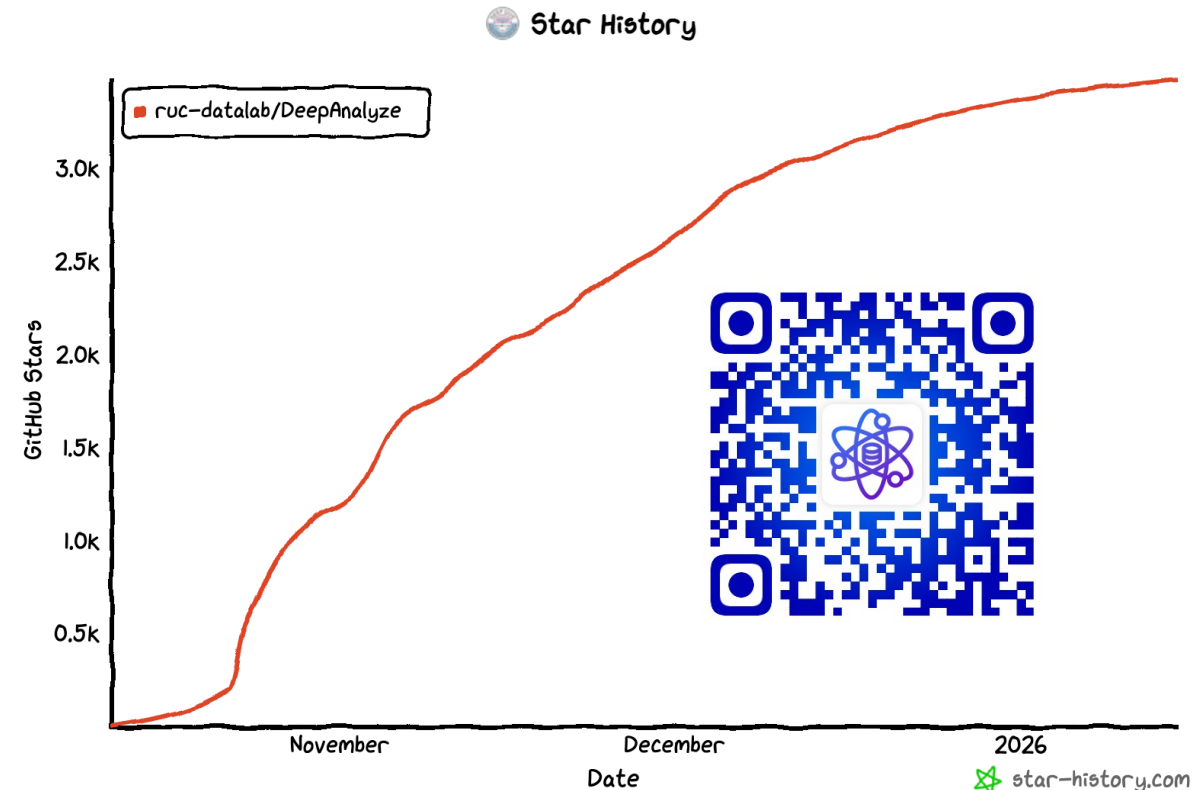
DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

Published on Oct 19 · Submitted by Shaolei Zhang on Oct 21 **#1 Paper of the day** RUC-DataLab

Authors: Shaolei Zhang, Ju Fan, Meihao Fan, Guoliang Li, Xiaoyong Du

DeepAnalyze

- RUC-DataLab/DeepAnalyze-8B
Text Generation · 8B · Updated 26 days ago · 6.1k · 67
- RUC-DataLab/DataScience-Instruct-500K
Viewer · Updated Oct 21 · 26.2k · 9.77k · 60



DeepAnalyze引起社区广泛讨论



Brian Roemmele @BrianRoemmele · 13h
BOOM!

Powerful scientific research now performed by your local AI is here!

Garage researchers with no pedigree can now lift humanity to new height of discovery and we ain't asking permission.

I have been testing **DeepAnalyze-8B**, an open-source agentic language model and am

[Show more](#)

Figure 1. DeepAnalyze-8B is the achieves autonomous data science pipeline and open-ended data res

25 74 3

Dr Singularity @Dr Singularity · 14h

Science is rapidly getting automated, and that means a 1000x acceleration is near.

DeepAnalyze-8B, the first agentic LLM designed for autonomous data science, capable of automatically completing the end to end pipeline from data sources to analyst grade deep research report.

DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

Shaolei Zhang¹ Ju Fan¹ Meihao Fan¹ Guoliang Li² Xiaoyong Du¹

[ruc-datalab/DeepAnalyze](#) [DeepAnalyze-8B](#) [DataScience-Instruct-500K](#) [ruc-deepanalyze.github.io](#)

Abstract

Autonomous data science, from raw data sources to analyst-grade deep research reports, has been a long-standing challenge, and is now becoming feasible with the emergence of powerful large language models (LLMs). Recent workflow-based data agents have shown promising results on specific data tasks but remain fundamentally limited

Oct 2025

26 77 480 30K

绍磊 Shaolei Zhang @Vily_1998

Welcome everyone to use and submit issues & pull requests — we're very happy to collaborate with you all in contributing to DeepAnalyze.

Python Developer @Python_Dv · 16h

DeepAnalyze: Agentic Large Language Models for Autonomous Data Science

DeepAnalyze is the first agentic LLM for autonomous data science. It can autonomously complete a wide range of data-centric tasks without human intervention, supporting:...

[Show more](#)

[Show this thread](#)

11:41 AM · Oct 22, 2025 · 28 Views

View post engagements

Post your reply

光影织梦 @SunMousekomre · 20m

We tested it. The fine-tuned model has an accuracy rate that is 20-30% higher than the agent analysis we did based on the general model.

1 4

DeepAnalyze-8B为自主数据科学提供了一种解决方案

Gorden Sun @Gorden_Sun · Oct 21

DeepAnalyze: 数据分析智能体

能自主完成一系列数据任务, 包括: 准备、分析、建模、可视化、生成报告。

Github: [github.com/ruc-datalab/De...](#)

3 69 219 16K

arXiv:2510.168

also introduce a data-grounded trajectory synthesis framework that constructs high-quality training data. Through agentic training, *DeepAnalyze* learns to perform a broad spectrum of data tasks, ranging from data question answering and specialized analytical tasks to open-ended data research. Experiments demonstrate that, with only 8B parameters, *DeepAnalyze* outperforms previous workflow-based agents built on most advanced proprietary LLMs. The model, code, and training data of *DeepAnalyze* are open-sourced, paving the way toward autonomous data science.

advances in large language models (LLMs) have demonstrated impressive problem-solving abilities (OpenAI, 2023; 2024; DeepSeek-AI, 2025), reshaping paradigms in domains such as search (Zheng et al., 2025; Jin et al., 2025) and mathematics (Zhang et al., 2024; Ren et al., 2025). However, despite their success on unstructured data (e.g., textual queries or contexts), LLMs still struggle to orchestrate complex, multi-stage data science pipelines and handle diverse structured data, making it difficult to achieve a general solution that works across all data science tasks.

Addressing these challenges requires endowing LLMs with two higher-level capabilities: autonomous orchestration and adaptive optimization. First, autonomous orchestration en-

Rohan Paul @rohanpaul_ai · 23h

This paper builds an agentic LLM that can run the whole data science workflow by itself.

It is an 8B model that plans work, reads structured files, writes and runs code, checks results, and iterate...

26 77 480 30K

社区反馈

DeepAnalyze在数据分析上显著优于通用模型上构建的智能体

光影织梦 @SunMousekomre · 20m

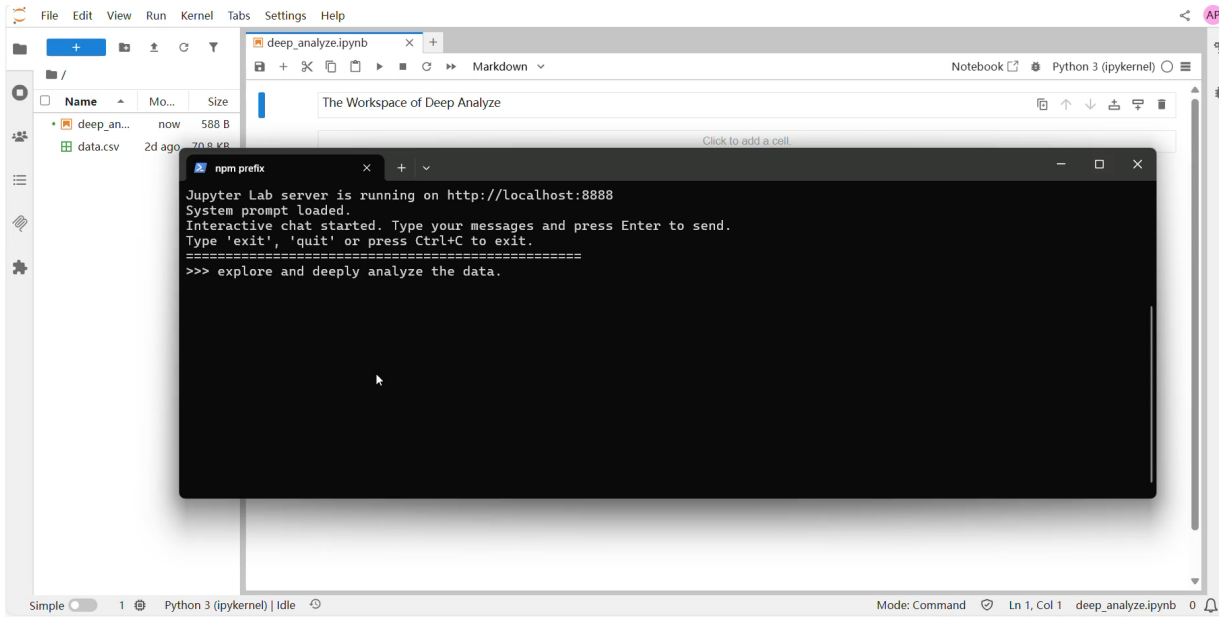
We tested it. The fine-tuned model has an accuracy rate that is 20-30% higher than the agent analysis we did based on the general model.

1 4

DeepAnalyze在社区实践中初具成效

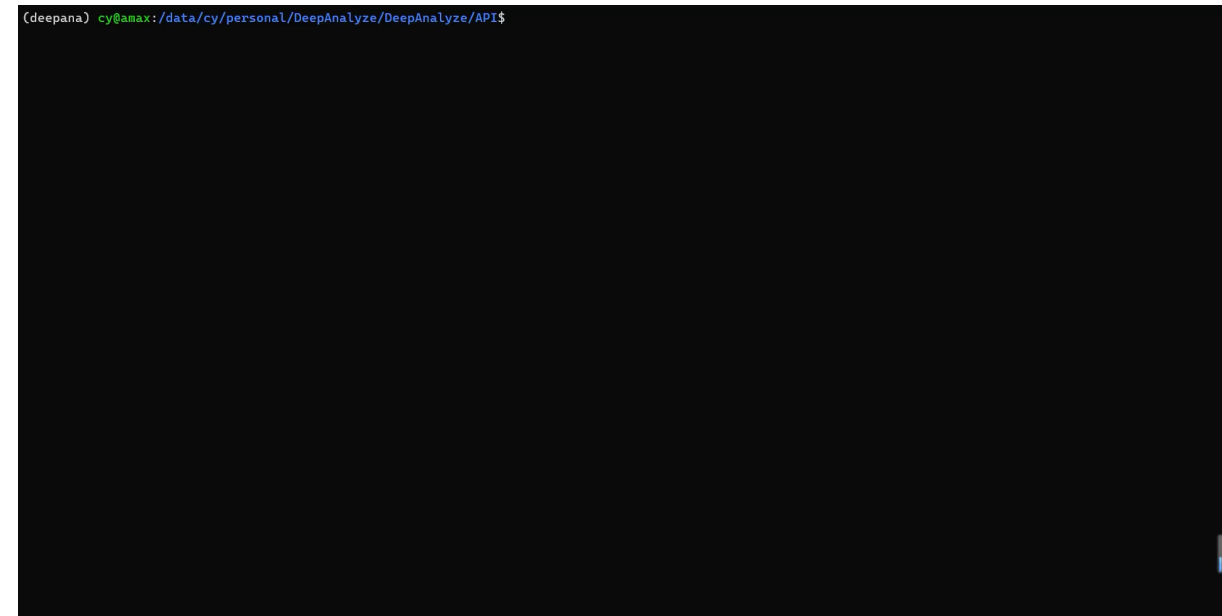
人机协同方式

通过WebUI、JupyterUI、CLI模式交互



JupyterUI

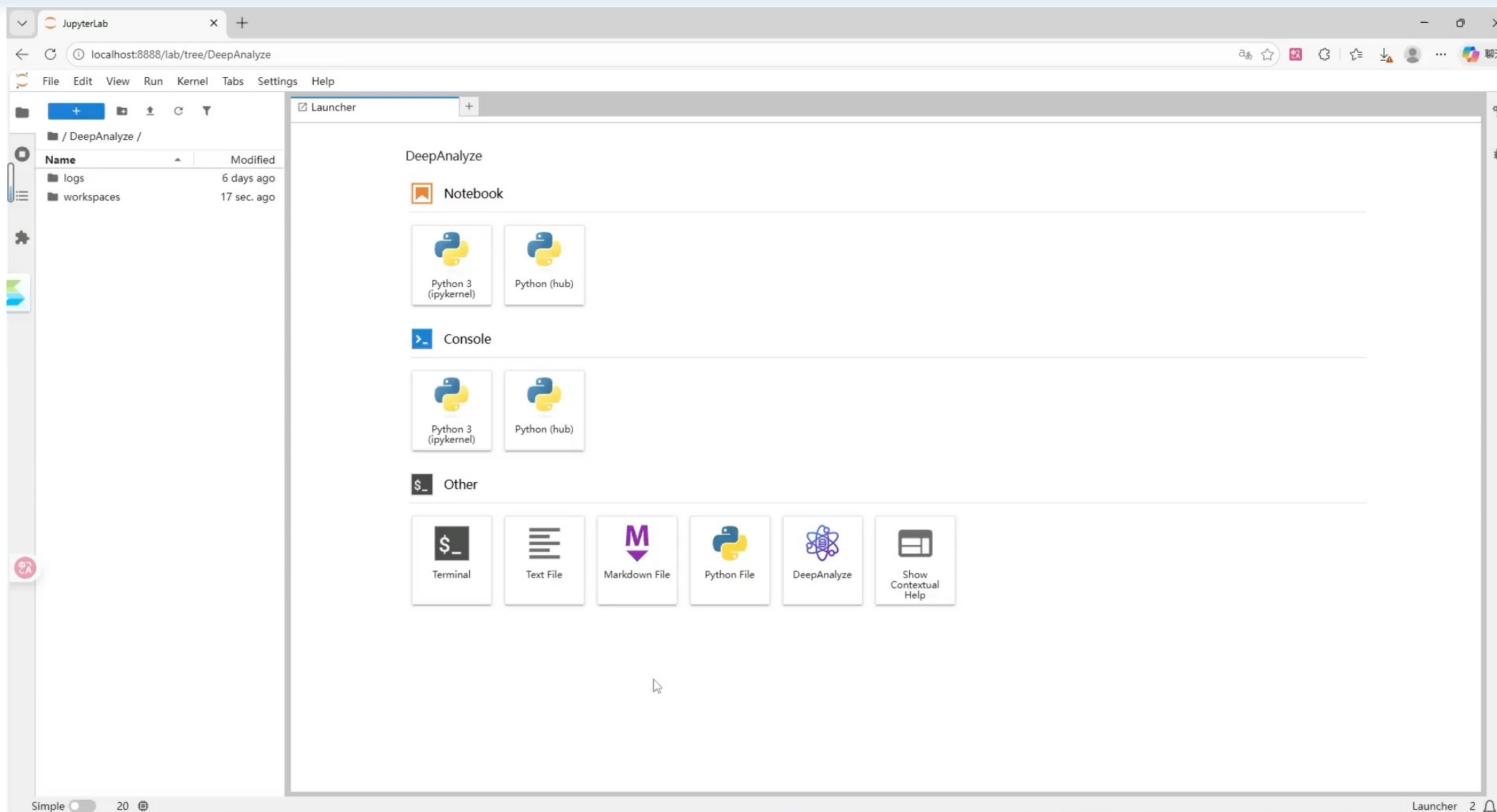
通过自然语言控制Jupyter notebook对数据进行分析



CLI

通过命令行交互对数据进行分析

Vibe Analyzing with DeepAnalyze





DeepAnalyze在社区实践中初具成效

DeepAnalyze × 和鲸社区 收到300+用户内测申请

数据科学分析智能体 - DeepAnalyze

上传多个文件并输入分析指令，系统将自动分析文件内容并生成详细报告。

[项目源地址: DeepAnalyze GitHub仓库](#)

上传文件

expense_data.csv192.2 KB

product_margin_data.csv455.0 B

sales_data.csv336.9 KB

分析指令

帮我分析这些数据文件，看一下公司今天的整体运营情况，并给我生成一份专业完整的公司运营分析报告。

高级设置

模型选择

deepanalyze-8b

温度参数

0.5

最大令牌数

32768

开始分析

清空所有

分析结果输出

文件分析系统 - 处理结果

基本信息:

- 分析指令: 帮我分析这些数据文件，看一下公司今天的整体运营情况，并给我生成一份专业完整的公司运营分析报告。
- 工作目录: /home/mw/project/temp_workspace
- 文件处理: 成功 3/3 个文件

文件详情:

- ✓ expense_data.csv (192.20 KB)
- ✓ product_margin_data.csv (455.00 B)
- ✓ sales_data.csv (336.95 KB)

分析结果

<Analyze>

为了全面分析公司今天的整体运营情况，我需要从三个数据文件中提取关键信息并进行整合分析。以下是初步的分析计划:

1. 数据初步探索:

- 首先分别检查每个文件的结构（列名、数据类型、缺失值）
- 了解每个文件的核心业务含义（费用、产品利润、销售数据）
- 检查时间维度是否一致（是否都是今日数据）

2. 关键指标识别:

- 从销售数据中提取销售额、销售量
- 从费用数据中提取各类运营成本
- 从产品利润数据中提取产品线利润贡献

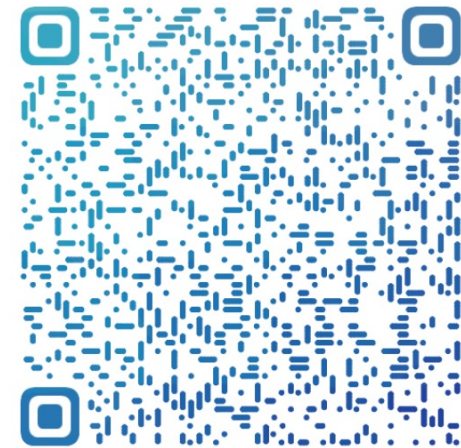
3. 整体运营分析框架:

- 收入分析（销售表现）
- 成本结构分析
- 利润能力分析
- 产品线表现对比
- 成本效益分析

4. 可视化需求:

- 销售额时间趋势（如小时级）
- 成本构成图

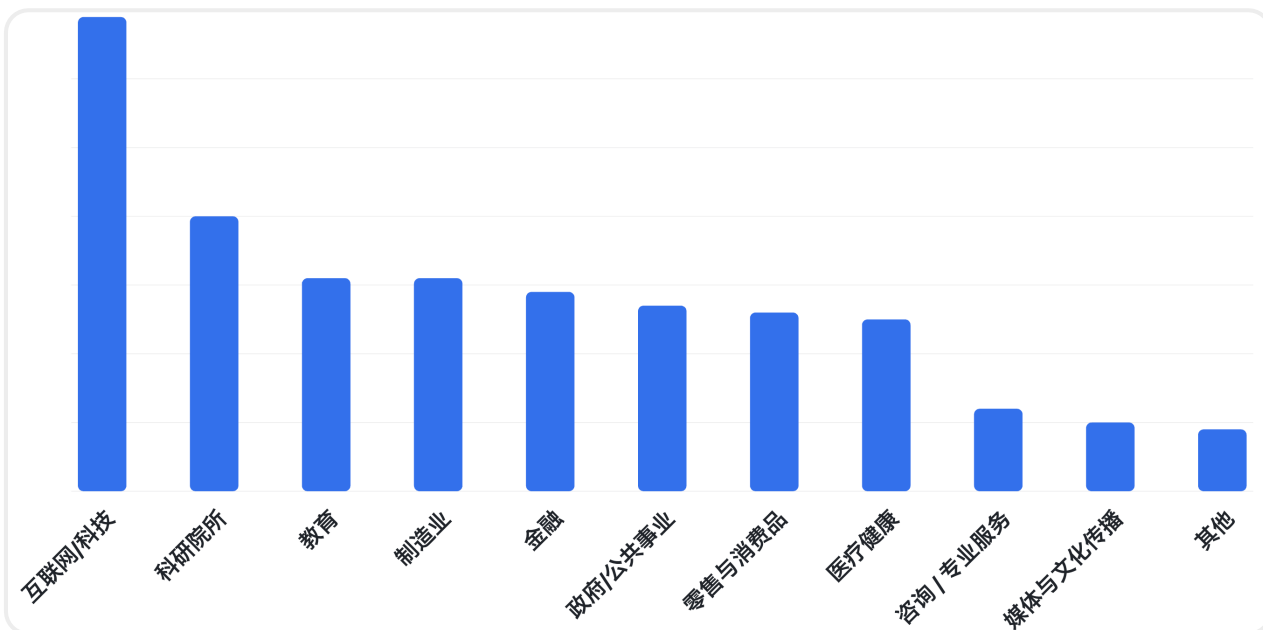
在ModelWhale在线体验DeepAnalyze



DeepAnalyze内
测开放申请中

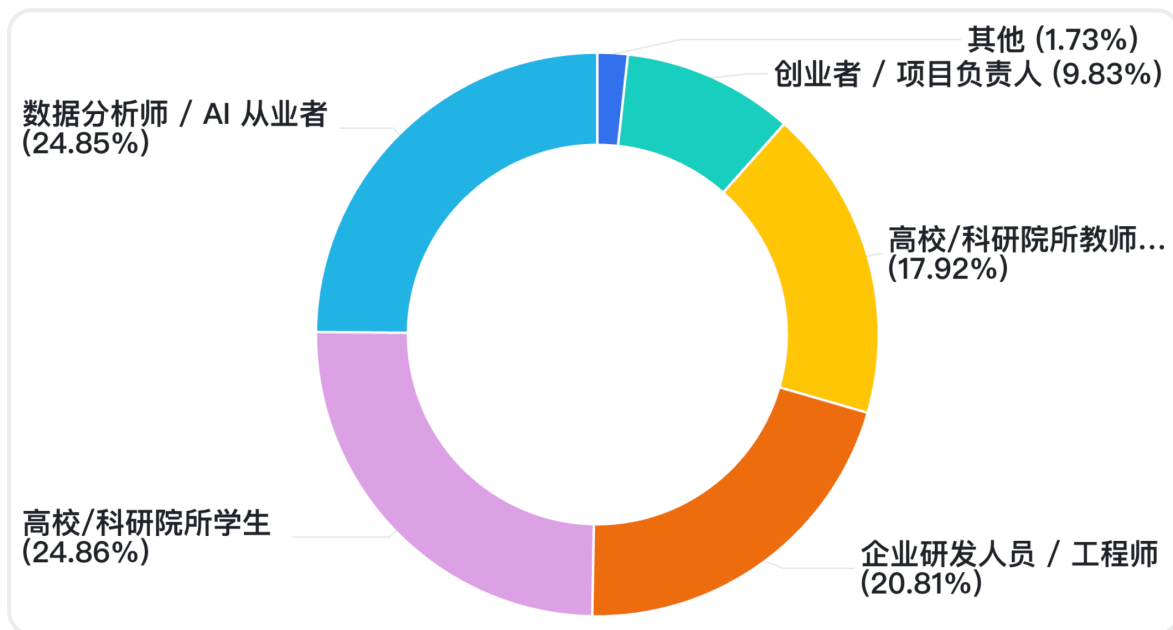
DeepAnalyze在社区实践中初具成效

DeepAnalyze × 和鲸 收到400+来自院校/企业的内测申请



跨领域应用潜力巨大

来自互联网、教育、金融、制造、医疗、政府、零售、咨询各领域



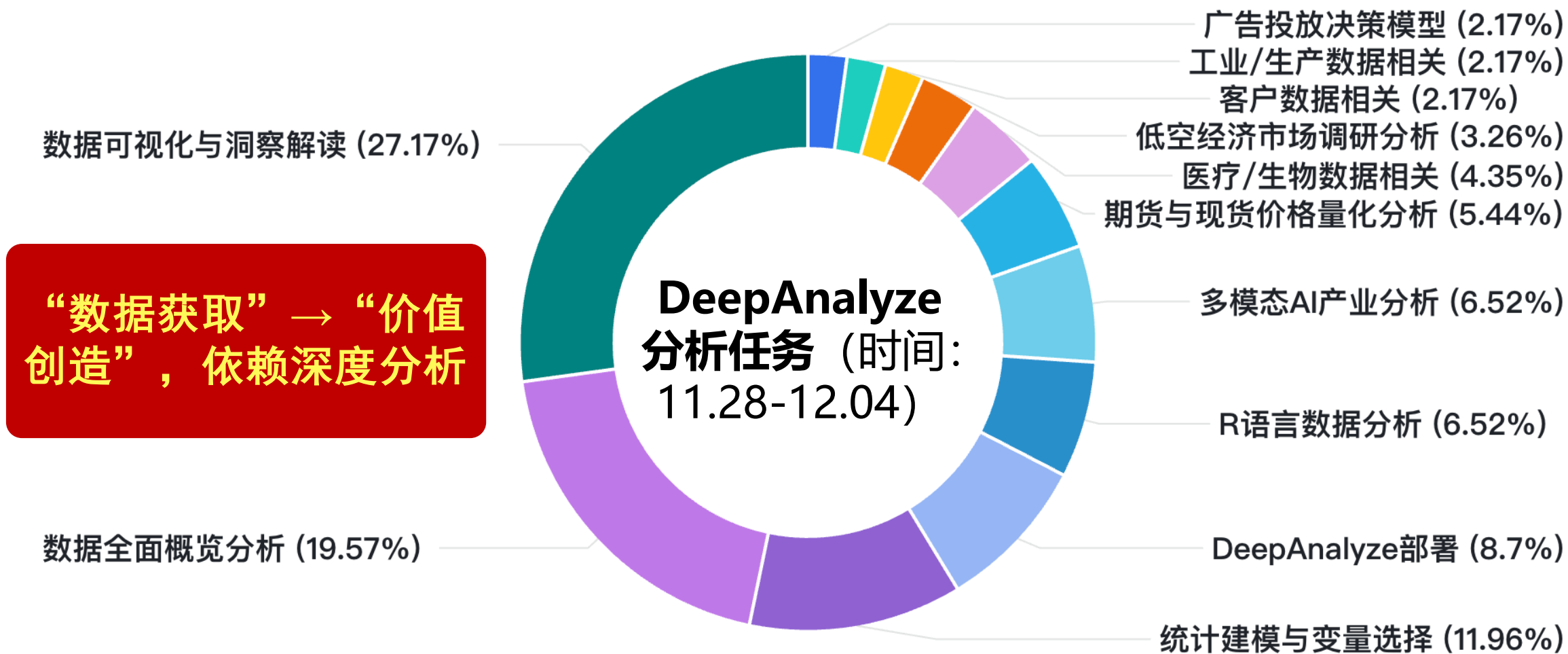
深度覆盖“研”与“用”群体

“研”：高校/科研院所教师、学生、研究员

“用”：数据分析师、AI从业者、创业者

DeepAnalyze在社区实践中初具成效

DeepAnalyze × 和鲸 内测交互日志，反映DeepAnalyze的使用场景



DeepAnalyze 首个面向数据科学的Agentic大模型

- 课程式Agentic训练 + Agentic数据合成
- DeepAnalyze作为数据科学领域的基座模型 → 直接使用、特定领域微调、搭建复杂 workflow

DeepAnalyze
讨论群



联系方式



中國人民大學
RENMIN UNIVERSITY OF CHINA



中国人民大学信息学院
SCHOOL OF INFORMATION RENMIN UNIVERSITY OF CHINA

谢谢大家



联系方式

张绍磊

中国人民大学

zhangshaolei98@ruc.edu.cn

<https://zhangshaolei1998.github.io/>



个人主页